Repository Appendix D: Detailed Results for "Benchmark-Gated Composite Mean Estimation: Regime-Conditional Improvement and Its No-Free-Lunch Limits"

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Relationship to the journal submission

This repository appendix accompanies the article and provides the longer result narrative, additional result tables, and supporting visualizations. Appendices A–C remain in the journal supplementary file because they contain the regime construction, audit evidence, and theoretical and methodological details required for peer review. The version-controlled project is available at `https://github.com/ALRIER/Genetic_Algo_Hull_University`, and the journal supplement should be consulted for Supplementary Tables S1–S7 and Supplementary Figures S1–S3. This document does not replace the journal supplementary material.

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Repository Appendix D: Detailed discovery, confirmation, audit, and external results

D.1 Initial discovery outcomes

The initial discovery stage produced a selective map of regime-dependent improvement under both discovery seeds. Under seed 101, genetic algorithm composites passed the dual benchmark gate in 9 of 18 family and regime cells; under seed 202, they passed in 7 of 18. Across both seeds, the procedure accepted composites in 16 of 36 seed-specific evaluations and retained the benchmark in the remaining 20.

The strongest family-level gains were concentrated in the Inverse Gaussian and Weibull families under both seeds. Inverse Gaussian produced three accepted cells under seed 101 and two under seed 202, with maximum $q_{0.95}$ MSE gains of 66.3% and 70.9%, respectively. Weibull produced two accepted cells under each seed, with corresponding maximum gains of 55.3% and 53.1%. Lognormal also produced two accepted cells under each seed, while Ex Wald declined from two accepted cells under seed 101 to one under seed 202. Normal and Ex Gaussian remained benchmark-dominant under both discovery seeds.

The individual gain columns in Table 1 of the main text must be read together with the dual gate. For example, Ex Gaussian produced a positive $q_{0.95}$ point estimate under seed 202, but mean MSE deteriorated and the candidate was rejected. A composite was counted as a discovery win only when both mean MSE and $q_{0.95}$ MSE improved simultaneously.

The accepted weight vectors remained directly interpretable. Depending on the regime, the composites concentrated their mass on familiar components such as the mean, median, 20% trimmed mean, geometric mean, Huber location, and Tukey biweight location. Figure D1 shows that the accepted candidates generally concentrated their weight on a limited subset of components. The discovery process therefore produced interpretable combinations of established location estimators without applying opaque transformations to the data.

D.2 Fixed-weight confirmation and seed stability

The fixed-weight evaluation was substantially more selective than the initial discovery stage. The 25 focal candidates were assessed using eight additional validation seeds, $R = 500$ Monte Carlo replicates, $B = 500$ paired

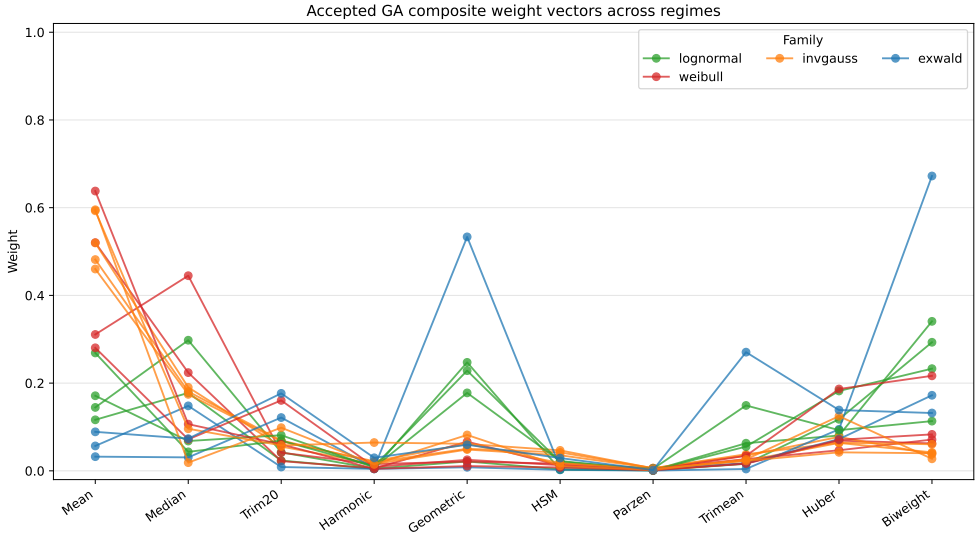


Figure D1: Accepted genetic algorithm weight vectors from the initial discovery stage. Each line represents one accepted composite estimator, and the axes correspond to the ten estimators in the initial convex basis.

bootstrap resamples, the expanded modern benchmark set, and both original and locked-unseen validation modes. Under this stronger comparison, transferable evidence was concentrated in the Lognormal family.

Three Lognormal candidates retained different levels of support (Table 2 of the main text). CV-019 and CV-010 passed the expanded gate in the locked-unseen mode under all eight validation seeds, and every seed-specific $q_{0.95}$ paired bootstrap interval was entirely above zero. CV-019 was the strongest result, with a mean $q_{0.95}$ MSE improvement of 16.3% and an envelope across the eight seed-specific intervals from 7.6% to 23.2%. CV-010 produced a smaller mean improvement of 2.2%, with a corresponding envelope from 1.6% to 3.1%. This smaller result is statistically confirmed under the study gate but operationally modest; it should not be read as an automatic recommendation to replace a simpler benchmark. Seed-specific gains, interval endpoints, and dual-gate outcomes are reported in Supplementary Table S2 in the journal supplementary material.

CV-011 differed from these two confirmed candidates. Its mean $q_{0.95}$ MSE improvement across the eight original-regime validation seeds was 3.8%. The range from 2.8% to 5.0% is the envelope obtained from the minimum lower and maximum upper endpoints of the eight seed-specific paired bootstrap

intervals. It is not a pooled confidence interval and was not calculated conditionally on the two successful seeds. All eight seeds showed a positive $q_{0.95}$ point gain, but the dual gate was cleared only under seeds 303 and 505 because mean MSE improved only under those two seeds. CV-011 did not retain its advantage in the locked-unseen mode and was interpreted as a local $q_{0.95}$ signal with no evidence of transferability. The complete eight-seed results are reported in Supplementary Table S2 in the journal supplementary material.

Figure 1 of the main text displays the seed-averaged $q_{0.95}$ gains and the envelopes of the seed-specific paired bootstrap intervals for these three signals. The envelopes for CV-019 and CV-010 remain entirely to the positive side of the zero-gain reference line, consistent with their eight-seed locked-unseen confirmation. CV-011 also has a positive $q_{0.95}$ envelope in the original regime, but this does not represent dual-gate confirmation: six seeds failed because their mean MSE gains were nonpositive.

The family-level decision matrix further demonstrates the selectivity of the expanded gate (Figure D2). Lognormal was the only family to produce a confirmed signal in the locked-unseen mode during this stage. Weibull, Inverse Gaussian, and Ex Wald, despite their initial discovery wins, reverted to the strongest modern benchmark when the original ten-component weights were frozen and evaluated under the expanded comparison.

Normal and Ex Gaussian continued to serve as negative controls. Neither family produced a locked-unseen pass, and both remained benchmark-dominant. Fixed-weight confirmation reduced the number of accepted candidates and showed that the initial discovery map did not hold under stronger validation.

The seed-level pattern provides a further distinction between transferable and local evidence. CV-019 and CV-010 passed the expanded gate under validation seeds 303, 404, 505, 606, 707, 808, 909, and 1010. Their survival across all eight seeds suggests that the confirmed result was not driven by an isolated random stream. CV-011 passed only under two seeds in the original mode, which supports its classification as a local candidate. The complete seed-by-candidate matrix is reported in Supplementary Figure S1 in the journal supplementary material.

The negative control pattern was equally stable. Normal and Ex Gaussian produced no genetic algorithm passes in the locked-unseen mode across the eight validation seeds. These abstentions show that the expanded gate did not systematically favor composites produced by the genetic algorithm. It

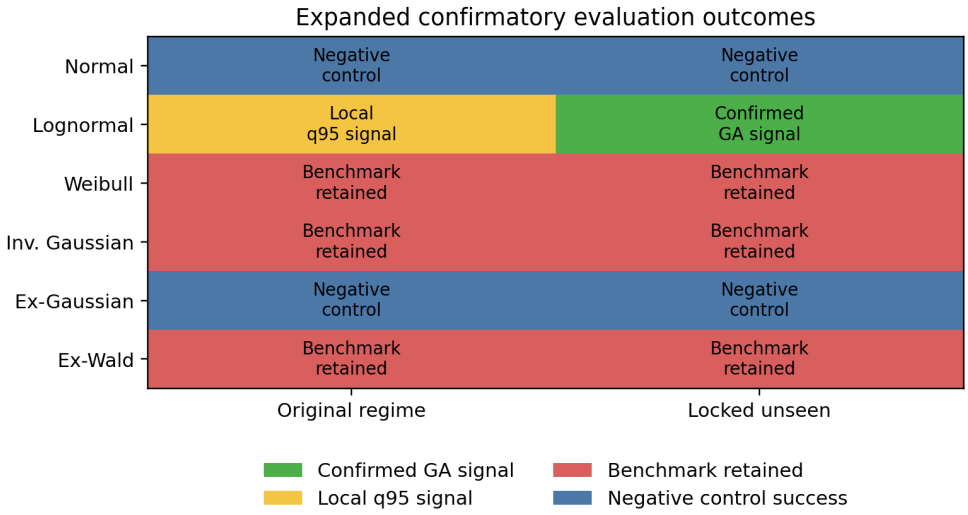


Figure D2: Fixed-weight confirmation outcomes by distributional family and validation mode. Green indicates locked-unseen confirmed genetic algorithm signal, yellow indicates a local original-regime $q_{0.95}$ signal, red indicates benchmark retention, and blue indicates negative-control success.

retained the benchmark when the modern estimator set was superior and accepted a composite only when its frozen weights continued to provide an advantage under stronger and previously unavailable conditions.

D.3 Expanded-basis rediscovery and transfer specialists

The expanded-basis search accepted 12 genetic algorithm candidates under the 26-component gate (Table D1). Accepted candidates emerged in five of the six distributional families: Ex Wald (3), Inverse Gaussian (4), Lognormal (2), Weibull (2), and Normal (1). Ex Gaussian remained benchmark-dominant. The largest discovery-stage $q_{0.95}$ MSE improvement appeared in the Inverse Gaussian family at 38.2%, followed by Weibull at 23.6% and Ex Wald at 20.6%.

The expanded estimator basis changed the family-level specialist map. Inverse Gaussian and Ex Wald produced the largest discovery gains, while the two Weibull candidates later provided the strongest evidence of transfer under fixed-weight validation. This result shows that introducing modern robust estimators directly into the learnable simplex opened candidate regions that were unavailable to the original ten-component search.

The confirmed Lognormal candidates from the initial stage, CV-019 and CV-010, entered the expanded search as controlled warm-start priors. They competed under the same 26-component gate as all other candidates and received no automatic acceptance. Lognormal still produced two accepted candidates, showing that the inherited weight structures remained competitive after Catoni-type, median-of-means, winsorized, and tuned Huber estimators became available within the search space.

Supplementary Table S7 in the journal supplementary material contrasts the two discovery stages, and Figure 2 of the main text shows the corresponding change in the family-level specialist map. The initial fixed-weight confirmation identified its strongest transferable signals in Lognormal regimes. The expanded-basis validation added two Weibull transfer specialists, extending the confirmed specialist set to a different distributional family and contamination structure.

Table D1: Expanded basis discovery outcomes by family under the 26-component genetic algorithm basis. Gains are discovery stage relative improvements before fixed-weight validation. Transfer specialist status was assigned only after the subsequent post-discovery fixed-weight validation.

Family	**Expanded stage GA wins**	**Confirmed prior inherited**	**Best $q_{0.95}$ gain**	**Source regimes**
Normal	1	No	7.0%	REG02/s202
Lognormal	2	Yes: CV-019, CV-010	3.9%	REG01/s101, REG03/s101
Weibull	2	No	23.6%	REG01/s101, REG02/s202
Inverse Gaussian	4	No	38.2%	REG01/s101, REG02/s101, REG02/s202, REG03/s202
Ex Gaussian	0	No	–	benchmark retained
Ex Wald	3	No	20.6%	REG01/s101, REG03/s101, REG03/s202
Total	**12**	**2 priors**	–	–

The expanded discovery universe contained 36 seed-specific family and regime cells: six families, three regime classes, and two discovery seeds. Twelve cells cleared the discovery gate and entered the fixed-weight panel as accepted GA winners. From the 24 cells that did not clear that gate, seven borderline near-gate cells and six benchmark-control cells were also selected for diagnostic fixed-weight evaluation. The resulting validation panel therefore contained 25 candidates: 12 discovery winners and 13 deliberately retained nonwinners. The remaining 11 benchmark-retained discovery cells were not included in this panel. The subsequent fixed-weight validation organized these 25 candidates into four evidence grades (Table D2). The taxonomy preserves the distinction between discovery success, diagnostic follow-up, and confirmed transfer.

Two Weibull candidates reached transfer specialist status after producing advantages supported by confidence intervals under locked-unseen conditions. Ten additional candidates retained discovery-supported local evidence: three Ex Wald, four Inverse Gaussian, two Lognormal, and one Normal candidate. These candidates identify valid discovery regions that remain relevant for further study, even though they did not meet the complete transfer standard.

Seven candidates were classified as near-gate cases. Their results approached the acceptance boundary and identify additional regions of the simplex that may benefit from greater discovery budgets or more targeted validation. Six Normal and Ex Gaussian cases served as negative-control successes, verifying that the expanded procedure continued to retain the benchmark when the composite did not provide sufficient improvement.

Figure 2 of the main text shows the basis-dependent change in the validated specialist map: the initial confirmation retained Lognormal signals, while the expanded basis added Weibull transfer specialists.

The four evidence grades were distributed selectively across families. Inverse Gaussian and Ex Wald contributed several discovery-supported candidates, confirming that the expanded basis found promising local regions in both families. Weibull produced the only two transfer specialists from this stage. Normal and Ex Gaussian also provided important control evidence by showing that basis expansion did not lead to indiscriminate composite acceptance. The complete family-level visualization is reported in Supplementary Figure S2 in the journal supplementary material. Candidate-level discovery gains, frozen locked-unseen gains, original-regime gains, and seed-level CI support for all 25 evaluated candidates are reported in Supplementary Table S3 in the journal supplementary material.

Table D2: Evidence taxonomy from the 25-candidate post-discovery fixed-weight validation panel ($N_{\mathrm{EST}} = 26$). The panel contains 12 accepted discovery winners, seven borderline near-gate cells, and six benchmark-control cells. The grades summarize the strength and scope of the available evidence without changing the discovery or validation gate decisions.

Evidence grade	**N**	**Families and interpretation**
transfer specialist	2	Weibull; confidence-interval-supported locked-unseen advantage
discovery-supported local candidate	10	Ex Wald (3), Inverse Gaussian (4), Lognormal (2), Normal (1)
near-gate candidate	7	Ex Gaussian (3), Normal (2), Weibull (2); approached the validation gate
negative-control success	6	Normal (3), Ex Gaussian (3); benchmark correctly retained

The two Weibull transfer specialists are reported in Table 3 of the main text. Q1R011, discovered in Weibull REG01 under seed 101, improved mean MSE by 2.79% and $q_{0.95}$ MSE by 2.72% under locked-unseen validation. Q1R012, discovered in Weibull REG02 under seed 202, achieved corresponding gains of 3.18% and 1.25%.

The contraction from discovery to frozen locked-unseen validation was substantial and must be evaluated separately for each candidate. Q1R011's $q_{0.95}$ gain declined from 7.32% at discovery to 2.72%, a reduction of 62.8%. Q1R012's $q_{0.95}$ gain declined from 23.61% to 1.25%, a reduction of 94.7%. The 23.6% family maximum in Table D1 belongs to Q1R012 and should not be used as the discovery value for Q1R011. These reductions support the separation of search and frozen validation and are consistent with substantial discovery-stage optimism. Because locked-unseen evaluation also changes the contamination profile, the reductions do not isolate a pure winner's-curse effect. The corresponding stage-specific quantities for every evidence category are shown in Supplementary Table S3 in the journal supplementary material.

Across the eight locked-unseen validation seeds, the criterion-specific comparators for these two specialists were Huber, tuned Huber, or winsorized estimators, with no Catoni-type variant serving as the comparator. Their confirmed gains therefore did not result from comparison with an unusually weak Catoni-type tuning.

These gains were obtained against the strongest expanded benchmark after the candidate weights had been frozen. Both specialists also satisfied the validation requirements across all eight additional seeds. Their performance indicates that the expanded search identified reproducible weight structures for severe Weibull contamination conditions, including upper-tail and point-mass contamination at scale 30.

The specialist advantage was specific to the locked-unseen conditions. Under the original-regime evaluation, Q1R011 recorded changes of -15.77% in mean MSE and -45.06% in $q_{0.95}$ MSE. Q1R012 recorded changes of -15.20% and -34.56%, respectively. These results define the applicability boundary of the two estimators: their frozen weights improve performance in the validated extreme transfer conditions, but their use requires regime matching and should not be extended automatically across all Weibull settings.

Figure D3 displays the contrast between the two validation modes. Both locked-unseen $q_{0.95}$ estimates are positive, showing a reproducible advantage under the severe transfer conditions. The negative original-regime estimates mark the boundary outside which the specialists should not be activated.

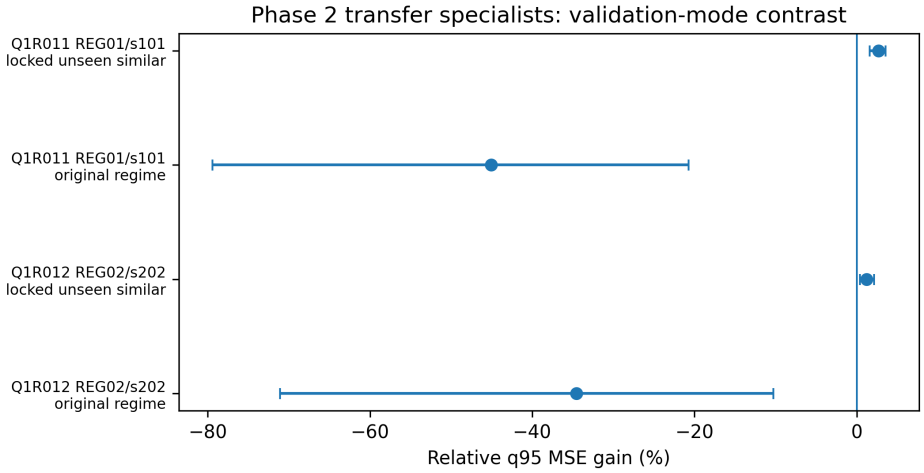


Figure D3: Validation mode contrast for the two Weibull transfer specialists. Points show $q_{0.95}$ MSE gains averaged across validation seeds, and intervals show the seed-level bootstrap confidence interval envelope across the eight validation seeds. Both specialists achieve positive gains under locked-unseen validation, while the original-regime results define the limits of their validated application.

D.4 Confirmation and abstention audits

The confirmation and abstention audits described in Section 3.4 of the main text preserved both Weibull specialist confirmations. No additional candidate met the complete confirmation standard.

The robustness audit showed that both specialists substantially reduced the influence of extreme observations. One arbitrarily large observation moved Q1R011 by approximately 1/62 of its effect on the sample mean and moved Q1R012 by approximately 1/46. Most of the frozen weight in both composites was assigned to procedures with bounded or attenuated influence, including Huber, Catoni-type, biweight, median, and winsorized estimators. The geometric and harmonic means together accounted for no more than 2.3% of the total weight.

Both specialists retained a small positive weight on the sample mean, with

$$w_{\text{mean}} = 0.016$$

for Q1R011 and

$$w_{\text{mean}} = 0.022$$

for Q1R012. Their formal breakdown point consequently remains zero. This theoretical limitation does not remove the observed reduction in gross-error sensitivity, which was one to two orders of magnitude below that of the sample mean. The corresponding frozen-weight audit is reported in Supplementary Table S4 in the journal supplementary material.

The bootstrap audit found that modal components contributed very little to either specialist. Their combined modal weights were 0.74% for the Weibull REG01 specialist and 0.43% for the Weibull REG02 specialist. The known limitations of the ordinary bootstrap for modal estimators are consequently unlikely to affect the confirmed intervals materially (Ghosh et al., 1984; Romano, 1988).

Median weight was more visible in selected candidates. It reached 17.9% in the Weibull REG01 specialist and 13.5% in the Lognormal REG01 candidate. These values justify continued caution when interpreting intervals containing a substantial median contribution. In both cases, however, the median formed part of a broader weighted estimator whose remaining mass was concentrated largely on smooth components with $\sqrt{n}$ consistency.

Multiplicity correction reinforced the stability of the specialist results. Both Weibull candidates survived the Benjamini–Hochberg correction at the 5% level across all eight validation seeds, and their $q_{0.95}$ lower confidence

bounds remained strictly above zero. The aggregate one-sided signal was at or below the bootstrap resolution,

$$1/B = 1/500$$

per seed. The correction preserved the two confirmed specialists and did not elevate any additional candidate.

The split-sample audit closely reproduced the pooled estimates. Q1R011 achieved a held-out $q_{0.95}$ MSE gain of 2.84%, compared with its pooled gain of 2.72%. Q1R012 achieved a held-out gain of 1.28%, compared with a pooled gain of 1.25%. The differences were only 0.12 and 0.03 percentage points, respectively, providing additional evidence that the reported gains were not driven by selection within the complete validation seed collection.

The validation seeds 303 through 1010 were distinct from the discovery seeds 101 and 202, adding an independent layer to confirmation. The earlier CV-019 Lognormal specialist remains the largest selected result on a structurally similar locked-unseen regime, with a $q_{0.95}$ MSE gain of 16.3% and a lower confidence bound of 7.6%. Because each half of the split-sample audit contains only four seeds, those estimates are used as robustness checks and not as separate inferential intervals.

The random Dirichlet audit also supported the selectivity of the benchmark gate. Among the 34 regime and validation mode cells, 23 confirmed abstention because none of the 4,000 sampled composites cleared the dual gate with stability across all eight seeds. Nine of the 17 audited regimes confirmed benchmark retention in both validation modes.

The audit successfully distinguished positive and negative controls. All six Normal and Ex Gaussian control regimes confirmed abstention; no random vector passed the gate across all eight seeds. The two frozen Weibull transfer specialists each passed under all eight seeds when included as positive controls. This contrast shows that the audit could recover established specialist signals while continuing to retain the benchmark in the designated control families. Complete audit counts are reported in Supplementary Table S5 in the journal supplementary material.

The Dirichlet audit also identified new configurations for future investigation. Eight heavy-tailed regimes contained at least one random composite that passed across all eight seeds but had not been recovered by the completed genetic algorithm run. These signals occurred among Ex Wald and Inverse Gaussian regimes, two Lognormal regimes, and one Weibull near-gate

regime. No corresponding signal appeared in the Normal or Ex Gaussian control families.

The strongest new search target was Q1R003 in Ex Wald, where 72 of the 4,000 random draws passed the stability requirement, corresponding to a 1.8% pass rate. Pass rates in the remaining flagged regimes were below 0.3%. These results show that gate-passing configurations can be uncommon under the selected Dirichlet proposal distribution and provide concrete targets for future persistence and frozen validation. They were not added to the confirmed set because the sampled weights had not yet completed the full validation procedure.

Taken together, the audits strengthened the main findings of the expanded stage. They preserved both Weibull transfer specialists, confirmed benchmark retention across the Normal and Ex Gaussian controls, reproduced the specialist gains under multiplicity and split-sample checks, and identified eight focused search targets for future studies. The resulting evidence supports two validated specialists while also providing a clear map for extending the discovery process.

D.5 External validation on public data

The external battery requested 264 public registry entries, successfully loaded 228, and selected 128 after balancing profile quotas and source caps. Eight selected entries were not scored because their operational profiles had no eligible specialist mapping: four were classified as zero-inflated/count and four as empirical-unknown. The remaining 120 entries produced dataset-level evaluations. Among them, 43 independent parent datasets satisfied the profile eligibility requirements for primary testing.

Twelve of the 25 frozen candidates achieved at least one strict external win. After repeated evaluations were collapsed to their independent parent datasets, the candidates produced wins confirmed by confidence intervals and surviving false discovery rate control on 26 of the 43 eligible parents. The extent of external transfer was measured across these 26 independent datasets, with lower-level comparisons excluded from the independent dataset count.

When strict wins were summarized by empirical profile, 14 parent datasets contributed to Lognormal, 12 to Normal, and 5 to Weibull. These profile-specific counts are not mutually exclusive because distinct marginals and depth probes from the same parent dataset could activate under different empirical profiles. They consequently do not sum to the 26 unique parent datasets used as the primary measure of independent dataset coverage. No

profile-eligible external win was observed in the Ex Gaussian, Ex Wald, or Inverse Gaussian families.

Evidence within the successful parent datasets was also extensive. Across eligible subregime comparisons, 256 rows reached the strong-win threshold supported by confidence intervals. Of these, 255 survived the Benjamini–Hochberg correction at the 5% level and 257 survived at the 10% level. The additional two rows at the 10% threshold included one strict strong win and one sensitivity-only row. Among the 255 confirmations at 5%, 203 were within-dataset depth probes and were not counted as additional independent datasets.

Figure 3 of the main text separates these two evidential dimensions. The left panel reports profile-specific parent dataset coverage, and the right panel reports corrected subregime confirmations within those datasets. The external signal was concentrated in the same three empirical profile families in which the frozen candidates activated, while the remaining families retained their corresponding benchmarks.

The primary gate remained highly selective. The audit-ready gate table contained 5,900 condition-level rows spanning multiple contamination conditions within each dataset, specialist, and validation-mode combination. Of these lower-level rows, 2,211 had a positive paired bootstrap interval and were labelled strong-win cells. They are repeated condition-level observations, not bootstrap replicates and not independent evidence. Aggregation of these rows produced 809 profile-eligible dataset, specialist, and mode comparisons. Among the 809 aggregated comparisons, 256 contained at least one strong win supported by a confidence interval. After multiplicity correction, 255 survived at a 5% false discovery rate and 257 survived at the 10% sensitivity threshold. The external conclusion did not depend on adopting the more permissive threshold; the complete gate funnel is reported in Supplementary Figure S3 in the journal supplementary material.

External survivors were observed in both the original and locked-unseen related validation modes for the Lognormal, Normal, and Weibull profiles. In each of these families, the lower confidence bound on the reported gain remained strictly above zero. The external signal was not confined to a single validation mode within the successful families.

Table 4 of the main text summarizes the principal external counts and distinguishes independent dataset-level evidence from repeated confirmations within those datasets.

These results reinforce the interpretation of conditional transfer and

match the pattern observed in the simulation analyses. The frozen candidates improved on the prespecified profile benchmarks in 26 of the 43 eligible independent parent datasets. These activation areas also produced 255 corrected confirmations across profile-eligible subregime comparisons. The first result establishes transfer across independent empirical datasets; the second shows repeated support within the profiles where activation occurred.

This selective pattern also aligns with the No Free Lunch framing developed throughout the article. Broad dominance across most public datasets would not fit a regime-dependent specialist interpretation. The external gains remained concentrated in empirical profiles aligned with the validated specialist configurations. When the empirical profile did not support candidate activation, the framework retained the benchmark.

The external evidence remains limited to the operational target and contamination definition used in the battery. Both were constructed from each dataset's own marginal distribution. The findings establish profile-conditional recovery of the full-sample mean from repeated subsamples, but they do not validate performance against an independently known population mean. Within this scope, the external battery reinforces the simulation findings: the frozen composites activate selectively, reproduce their advantage under matched empirical conditions, and abstain outside the profiles supported by the evidence.

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